

WEEK 1 — Backend Foundation & Infrastructure

W1-BE-01 — Backend Project Initialization

Assignee: Mady

Deadline: End of Week 1

Description

Initialize the backend project with a clean, scalable architecture that will support all LMS features throughout the project lifecycle.

Acceptance Criteria

- Project initialized using NestJS
- Folder structure follows feature-based modular architecture
- Environment variables configured
- Global exception filter implemented
- Application runs successfully with a health check endpoint

Tech Stack

- Node.js
- NestJS
- TypeScript

Resources

- NestJS documentation
- Project ERD
- ESLint + Prettier

W1-BE-02 — Database Schema Implementation

Assignee: Mo

Deadline: End of Week 1

Description

Translate the approved ERD into a fully functional PostgreSQL database schema.

Acceptance Criteria

- All tables created exactly as ERD

- Primary & foreign keys enforced
- Enum types implemented
- Cascade rules defined
- Initial seed data created
- Migration scripts added

Tech Stack

- PostgreSQL
- TypeORM / Prisma

Resources

- ERD diagram
- pgAdmin
- Database normalization rules

W1-BE-03 — Authentication Core

Assignee: Kiro

Deadline: End of Week 1

Description

Implement secure authentication mechanisms for system users.

Acceptance Criteria

- Register endpoint implemented
- Login endpoint implemented
- Password hashing using bcrypt
- JWT issued on login
- Token validation middleware added
- Protected route successfully tested

Tech Stack

- JWT
- bcrypt
- NestJS Guards

Resources

- OWASP authentication guidelines
-

W1-BE-04 — Role-Based Access Control (RBAC)

Assignee: Boda

Deadline: End of Week 1

Description

Restrict system access based on user roles (Student, TA, Doctor, Admin).

Acceptance Criteria

- Role field stored in database
- Role-based guard implemented
- Unauthorized access blocked
- Roles tested on protected endpoints

Tech Stack

- NestJS Guards
- Custom decorators

Resources

- RBAC design patterns
-

W1-BE-05 — API Documentation Setup

Assignee: Amine

Deadline: End of Week 1

Description

Prepare API documentation to support team testing and frontend integration.

Acceptance Criteria

- Swagger enabled
- Auth endpoints documented
- Example requests/responses added
- Shared Postman collection created

Tech Stack

- Swagger
- Postman

Resources

- OpenAPI specification
-

WEEK 2 — Users & Courses Core

W2-BE-01 — User Profile Management

Assignee: Mady

Deadline: End of Week 2

Description

Enable users to manage their personal information.

Acceptance Criteria

- View profile endpoint
- Update profile endpoint
- Change password endpoint
- Input validation enforced
- JWT protection enabled

Tech Stack

- NestJS
- DTO validation

Resources

- User entity schema
-

W2-BE-02 — Course CRUD Module

Assignee: Mo

Deadline: End of Week 2

Description

Implement full lifecycle management for courses.

Acceptance Criteria

- Create course (Admin/Doctor only)
- Update course
- Delete course
- Fetch course details
- Role restrictions enforced

Tech Stack

- NestJS
- PostgreSQL

Resources

- Course table schema
-

W2-BE-03 — Enrollment System

Assignee: Kiro

Deadline: End of Week 2

Description

Allow students to enroll in available courses.

Acceptance Criteria

- Enrollment endpoint implemented
- Duplicate enrollment prevented
- Enrollment linked to user & course
- Access validation enforced

Tech Stack

- NestJS
- PostgreSQL

Resources

- Enrollment relation design
-

W2-BE-04 — File Upload Service

Assignee: Boda

Deadline: End of Week 2

Description

Enable file uploads for course materials and submissions.

Acceptance Criteria

- Upload endpoint implemented
- File metadata saved in DB
- File size/type validation
- Secure access implemented

Tech Stack

- Multer
- Local file system

Resources

- File upload best practices
-

W2-BE-05 — Notification System (Basic)

Assignee: Amine

Deadline: End of Week 2

Description

Implement basic notification functionality.

Acceptance Criteria

- Create notification endpoint
- Fetch notifications per user
- Mark notification as read

Tech Stack

- NestJS
- PostgreSQL

Resources

- Notification entity design
-

WEEK 3 — Academic Structure

(Modules, lessons, content, attendance, GPA)

W3-BE-01 — Module & Lesson Management

Assignee: Mady

Deadline: End of Week 3

Description

Organize courses into modules and lessons.

Acceptance Criteria

- Create module endpoint
- Create lesson endpoint
- Retrieve lessons per course
- Order preserved

Tech Stack

- NestJS
- PostgreSQL

Resources

- Academic hierarchy design
-

W3-BE-02 — Educational Content System

Assignee: Mo

Deadline: End of Week 3

Description

Attach educational resources to lessons.

Acceptance Criteria

- Content upload supported
- Content linked to lessons

- Multiple content types allowed

Tech Stack

- File service
- PostgreSQL

Resources

- LMS content models
-

W3-BE-03 — Course Access Validation

Assignee: Kiro

Deadline: End of Week 3

Description

Restrict lesson access to enrolled students only.

Acceptance Criteria

- Enrollment checked before access
- Unauthorized access blocked
- Proper HTTP errors returned

Tech Stack

- Guards
- Middleware

Resources

- Access control patterns
-

W3-BE-04 — Attendance Tracking

Assignee: Boda

Deadline: End of Week 3

Description

Track student attendance per course.

Acceptance Criteria

- Attendance record endpoint

- Fetch attendance history
- Data stored per user/course

Tech Stack

- NestJS
- PostgreSQL

Resources

- Attendance models

W3-BE-05 — GPA Calculation Logic

Assignee: Amine

Deadline: End of Week 3

Description

Automatically calculate student GPA.

Acceptance Criteria

- GPA calculated from graded submissions
- Stored in database
- Recalculated on grade update

Tech Stack

- Business logic layer

Resources

- University GPA formulas

WEEK 4 — Assignments System

W4-BE-01 — Assignment Management Module

Assignee: Mady

Deadline: End of Week 4

Description

Implement full CRUD operations for assignments linked to courses and modules.

Acceptance Criteria

- Create assignment endpoint (Doctor/TA only)
- Update assignment details
- Delete assignment
- Retrieve assignments per course
- Due date validation enforced

Tech Stack

- NestJS
- PostgreSQL

Resources

- Assignment table schema
- LMS assignment workflows

W4-BE-02 — Assignment Submission System

Assignee: Mo

Deadline: End of Week 4

Description

Allow students to submit assignments and track submission states.

Acceptance Criteria

- Submit assignment endpoint
- Prevent submission after deadline (unless allowed)
- Submission linked to user & assignment

- Submission status stored (submitted / graded)

Tech Stack

- NestJS
- PostgreSQL

Resources

- Submission entity design
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W4-BE-03 — Submission Attachments Handling

Assignee: Kiro

Deadline: End of Week 4

Description

Enable file attachments for assignment submissions.

Acceptance Criteria

- Upload files for submissions
- Validate file size & type
- Store file path in DB
- Files retrievable securely

Tech Stack

- Multer
- Local file system

Resources

- File upload security guidelines
-

W4-BE-04 — Assignment Grading System

Assignee: Boda

Deadline: End of Week 4

Description

Allow Doctor/TA to grade assignment submissions.

Acceptance Criteria

- Grade submission endpoint
- Score validation
- Feedback stored
- Grade visible to student

Tech Stack

- NestJS
- PostgreSQL

Resources

- Grading workflows
-

W4-BE-05 — Late Submission Rules

Assignee: Amine

Deadline: End of Week 4

Description

Apply late submission penalties automatically.

Acceptance Criteria

- Detect late submissions
- Apply penalty percentage
- Penalty configurable per course

Tech Stack

- Business logic layer

Resources

- Academic grading policies
-

WEEK 5 — Quiz System

W5-BE-01 — Quiz Management Module

Assignee: Mady

Deadline: End of Week 5

Description

Create and manage quizzes linked to courses.

Acceptance Criteria

- Create/update/delete quiz
- Time limit stored
- Attempt limit enforced

Tech Stack

- NestJS
- PostgreSQL

Resources

- Quiz entity schema
-

W5-BE-02 — Questions & Options Module

Assignee: Mo

Deadline: End of Week 5

Description

Manage quiz questions and answer options.

Acceptance Criteria

- Multiple question types (MCQ / True-False)
- Correct answer flagged
- Questions linked to quiz

Tech Stack

- NestJS
- PostgreSQL

Resources

- Online assessment models
-

W5-BE-03 — Quiz Attempt System

Assignee: Kiro

Deadline: End of Week 5

Description

Track quiz attempts per student.

Acceptance Criteria

- Start quiz attempt
- Track time spent
- Prevent extra attempts

Tech Stack

- NestJS
- PostgreSQL

Resources

- Quiz state machine logic
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W5-BE-04 — Auto Grading Logic

Assignee: Boda

Deadline: End of Week 5

Description

Automatically grade objective quiz questions.

Acceptance Criteria

- Auto-grade MCQ/TF
- Store score immediately
- Manual grading allowed if needed

Tech Stack

- Business logic services

Resources

- Auto-grading algorithms
-

W5-BE-05 — Quiz Results & Analytics

Assignee: Amine

Deadline: End of Week 5

Description

Provide quiz performance statistics.

Acceptance Criteria

- Average score calculated
- Pass/fail logic applied
- Results retrievable per quiz

Tech Stack

- PostgreSQL queries

Resources

- Educational analytics basics
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WEEK 6 — Discussion Forum

W6-BE-01 — Discussion Thread Management

Assignee: Mady

Deadline: End of Week 6

Description

Create discussion threads per course.

Acceptance Criteria

- Create discussion
- Edit discussion
- Delete discussion

Tech Stack

- NestJS
- PostgreSQL

Resources

- Forum design models
-

W6-BE-02 — Post System

Assignee: Mo

Deadline: End of Week 6

Description

Allow users to post inside discussions.

Acceptance Criteria

- Create post
- Edit post
- Delete post

Tech Stack

- NestJS
- PostgreSQL

Resources

- Forum posting logic
-

W6-BE-03 — Comment & Reply System

Assignee: Kiro

Deadline: End of Week 6

Description

Enable nested comments and replies.

Acceptance Criteria

- Comment on post
- Reply to comment
- Maintain hierarchy

Tech Stack

- PostgreSQL self-relations

Resources

- Threaded comment models

W6-BE-04 — Voting System

Assignee: Boda

Deadline: End of Week 6

Description

Implement upvote/downvote for posts.

Acceptance Criteria

- One vote per user
- Vote count updated correctly

Tech Stack

- NestJS
- PostgreSQL

Resources

- Voting system patterns

W6-BE-05 — Mark Answer as Solution

Assignee: Amine

Deadline: End of Week 6

Description

Allow Doctor/TA to mark best answers.

Acceptance Criteria

- Only authorized roles allowed
- Solution flag stored
- Highlighted in responses

Tech Stack

- NestJS guards

Resources

- Q&A forum best practices

WEEK 7 — Admin & System Control

W7-BE-01 — Admin User Management

Assignee: Mady

Deadline: End of Week 7

Description

Admin control over system users.

Acceptance Criteria

- Activate/deactivate users
- Assign roles

Tech Stack

- NestJS
- PostgreSQL

Resources

- Admin panel requirements

W7-BE-02 — Course Administration Controls

Assignee: Mo

Deadline: End of Week 7

Description

Administrative controls over courses.

Acceptance Criteria

- Archive course
- Reopen course

Tech Stack

- NestJS

Resources

- Course lifecycle rules

W7-BE-03 — Announcement System

Assignee: Kiro

Deadline: End of Week 7

Description

Broadcast announcements to users.

Acceptance Criteria

- Create announcement
- Target specific roles

Tech Stack

- NestJS

Resources

- Notification broadcasting logic

W7-BE-04 — System Logging & Auditing

Assignee: Boda

Deadline: End of Week 7

Description

Track important system actions.

Acceptance Criteria

- Log user actions
- Store logs securely

Tech Stack

- Winston Logger

Resources

- Audit logging standards

W7-BE-05 — Performance Optimization

Assignee: Amine

Deadline: End of Week 7

Description

Improve backend performance.

Acceptance Criteria

- Add DB indexes
- Optimize heavy queries

Tech Stack

- PostgreSQL

Resources

- SQL optimization techniques
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WEEK 8 — Security, Testing & Finalization

W8-BE-01 — Code Refactoring & Cleanup

Assignee: Mady

Deadline: End of Week 8

Description

Refactor codebase for maintainability.

Acceptance Criteria

- Remove unused code
- Improve naming & structure

Tech Stack

- TypeScript

Resources

- Clean code principles
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W8-BE-02 — Database Optimization

Assignee: Mo

Deadline: End of Week 8

Description

Finalize DB performance tuning.

Acceptance Criteria

- Index review
- Query plan analysis

Tech Stack

- PostgreSQL

Resources

- EXPLAIN ANALYZE
-

W8-BE-03 — Security Hardening

Assignee: Kiro

Deadline: End of Week 8

Description

Secure the backend against common attacks.

Acceptance Criteria

- Rate limiting
- Input sanitization
- CORS configured

Tech Stack

- Helmet
- Rate-limit middleware

Resources

- OWASP Top 10
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W8-BE-04 — Global Error Handling & Validation Review

Assignee: Boda

Deadline: End of Week 8

Description

Ensure consistent error responses.

Acceptance Criteria

- Unified error format
- All inputs validated

Tech Stack

- NestJS pipes

Resources

- API error standards
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W8-BE-05 — Final Documentation & Testing

Assignee: Amine

Deadline: End of Week 8

Description

Prepare system for submission & defense.

Acceptance Criteria

- Full API documentation
- All endpoints tested
- No critical bugs

Tech Stack

- Swagger
- Postman

Resources

- Graduation project checklist